

Project 4 — Healthcare Patient Analysis

Dataset note: All records are simulated for portfolio practice and are not suitable for clinical use.

Executive Summary

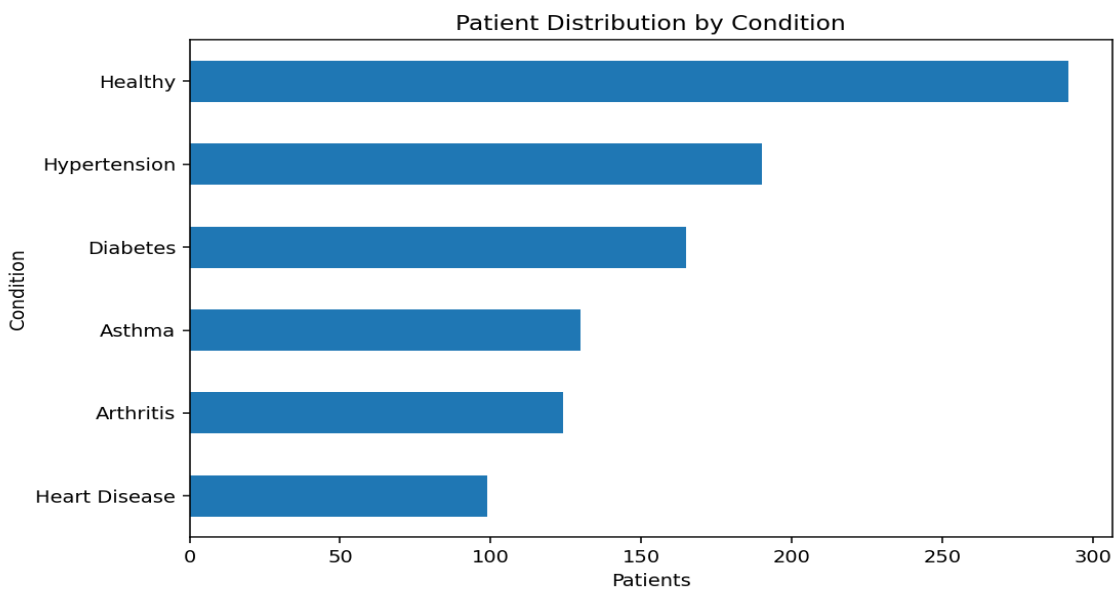
The cleaned dataset contains 1,000 synthetic patient records. Average age is 51.7 years, average BMI is 26.7, and average annual visits are 4.35. The synthetic high-risk label occurs in 30.2% of records.

Data Quality

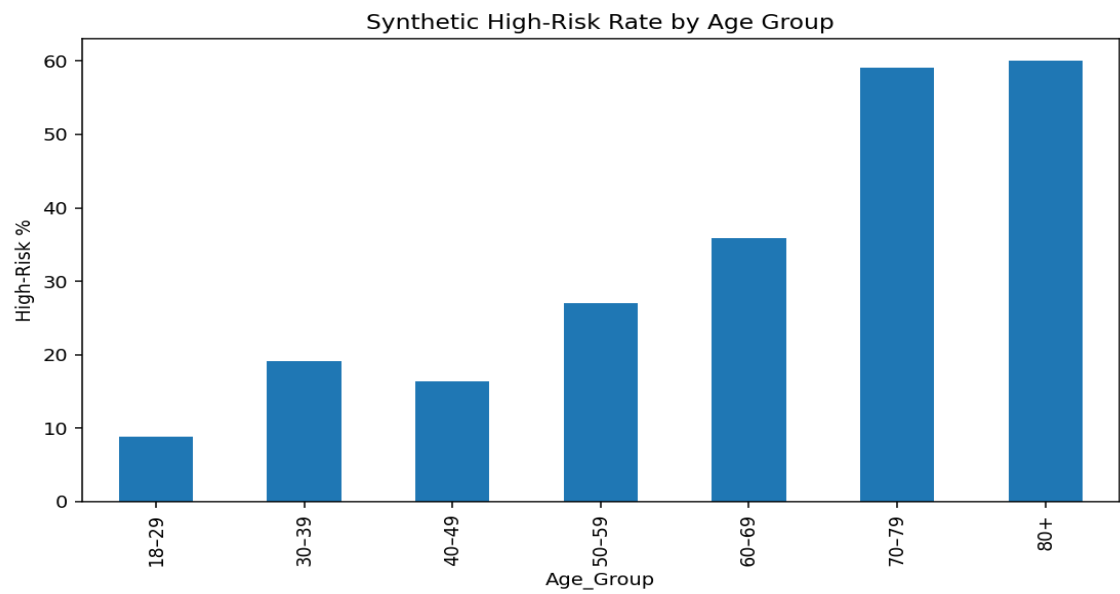
Metric	Result
Raw rows	1,002
Clean rows	1,000
Duplicates removed	2
Missing numeric fields	Median imputation

Condition Summary

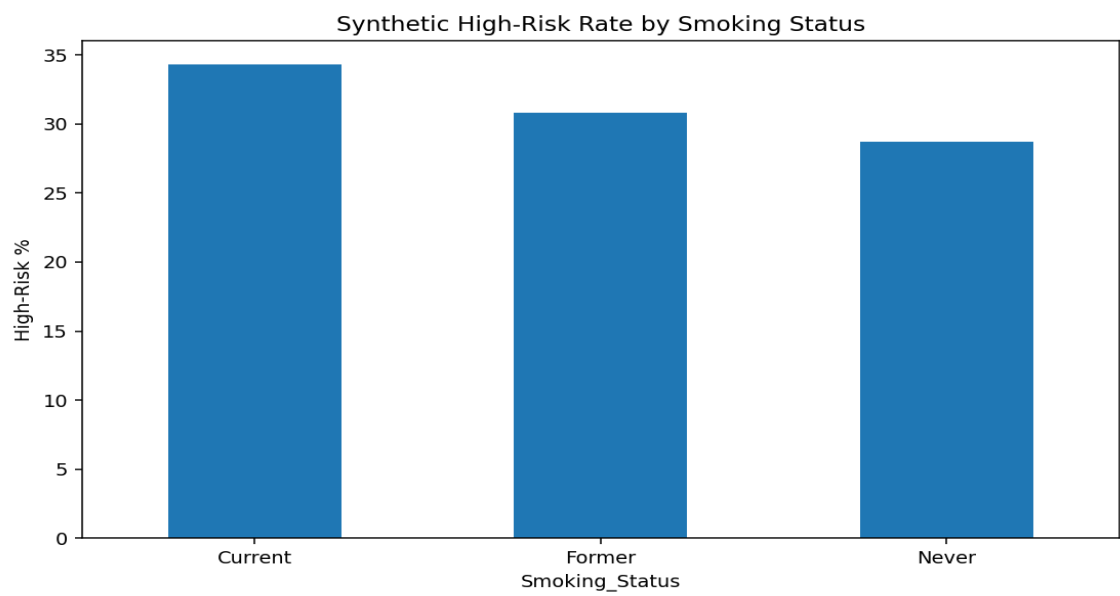
Condition	Patients	Avg Age	Avg Visits	Risk %
Healthy	292	51.3	3.80	26.4%
Hypertension	190	51.4	4.54	28.9%
Diabetes	165	53.7	4.63	39.4%
Asthma	130	51.3	4.61	23.8%
Arthritis	124	52.5	4.62	34.7%
Heart Disease	99	49.7	4.43	31.3%



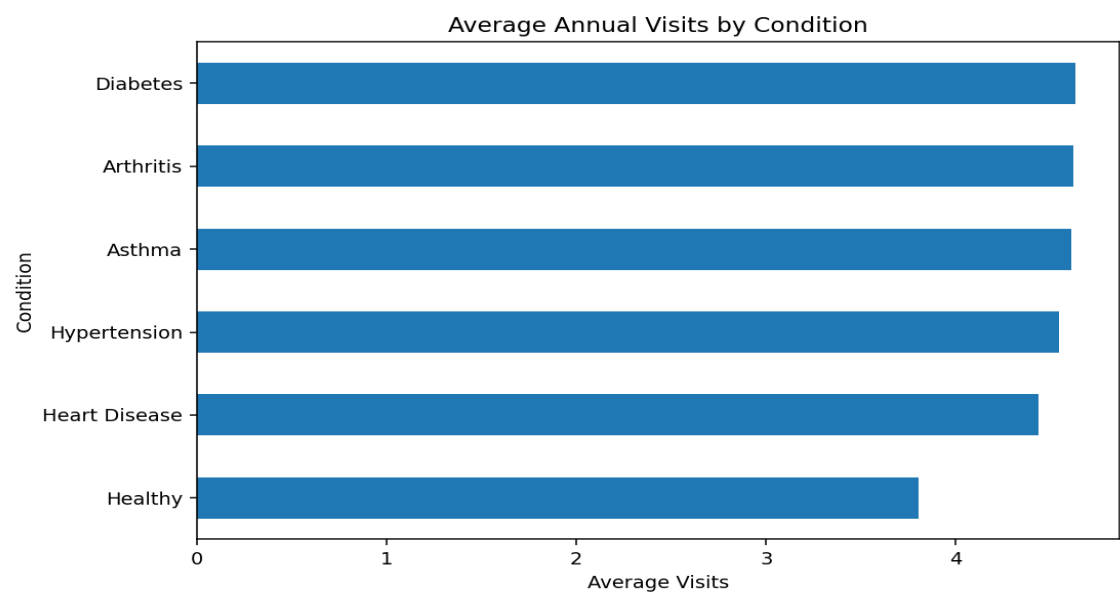
Patient distribution by condition



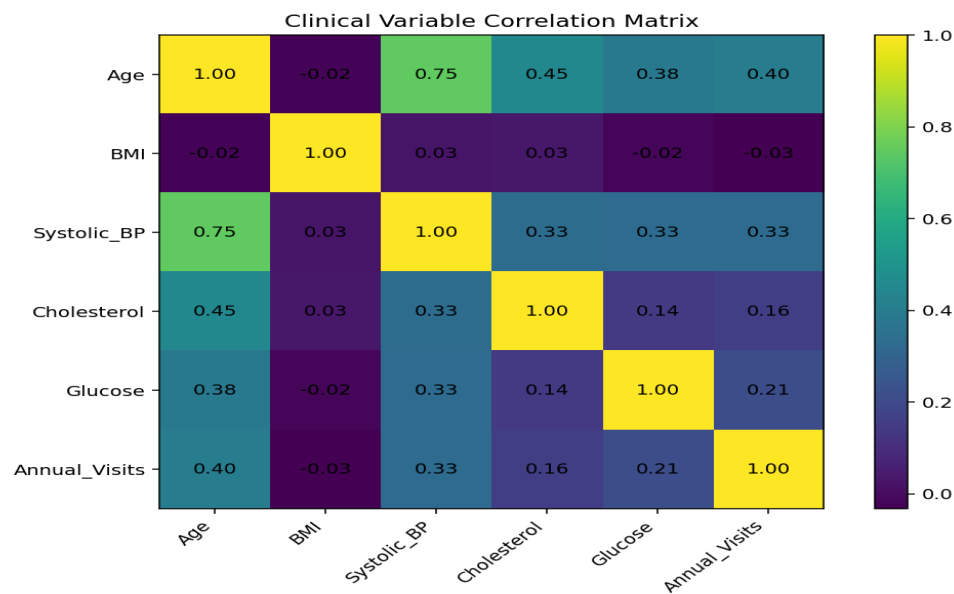
Synthetic high-risk rate by age group



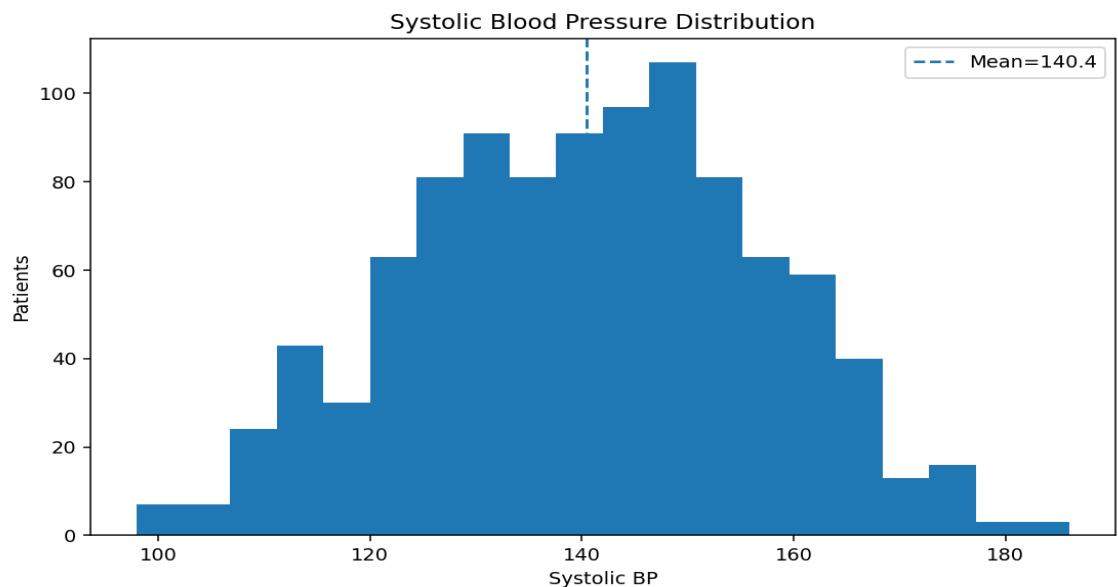
Synthetic high-risk rate by smoking status



Average annual visits by condition



Clinical-variable correlation matrix



Systolic blood pressure distribution

Insights and Responsible Interpretation

Condition, age, smoking status and visit patterns provide useful examples of healthcare segmentation. Correlations can identify relationships for further investigation but do not prove causation. Because every observation is synthetic, none of these results should be interpreted as medical evidence or generalized to real patients.